

Definition 2.2 (balls). Let (X, d) be a metric space. The *open ball* centered at a point $x_0 \in X$ with radius $r > 0$, denoted $B_r(x_0)$, is

$$B_r(x_0) = \{x \in X \mid d(x_0, x) < r\}.$$

Replacing $<$ by $\leq$ yields the *closed ball* $\overline{B}_r(x_0)$.

Definition 2.3 (open sets in metric spaces). A set $U \subseteq X$ is *open* in a metric space (X, d) if for every $x \in U$ there is some $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq U$.

Definition 2.4 (neighborhoods). A *neighborhood* of a point x in a metric space is an open set U containing it, $x \in U$.

Remark 2.1. Some authors call any set that merely *contains* an open set around x a neighborhood of x ; in these notes neighborhoods are always open.

Lemma 2.1. *Every open ball is an open set.*

Proof. Let $y \in B_r(x_0)$ and set $\varepsilon = r - d(x_0, y) > 0$. If $z \in B_\varepsilon(y)$, then the triangle inequality gives

$$d(x_0, z) \leq d(x_0, y) + d(y, z) < d(x_0, y) + \varepsilon = r,$$

so $B_\varepsilon(y) \subseteq B_r(x_0)$. Hence every point of $B_r(x_0)$ has a ball around it contained in $B_r(x_0)$, which is therefore open. $\square$

Proposition 2.1. *A set in a metric space is open iff it is a union of open balls.*

Proof. If $U = \bigcup_\alpha B_\alpha$ is a union of open balls and $x \in U$, then $x \in B_\alpha$ for some α , and since B_α is open (lemma 2.1) some ball around x lies in $B_\alpha \subseteq U$; hence U is open. Conversely, if U is open, then for every $x \in U$ choose a ball B_x with $x \in B_x \subseteq U$; then $U = \bigcup_{x \in U} B_x$ is a union of open balls. $\square$

Example 2.2. The open balls of $(\mathbb{R}, |\cdot|)$ are exactly the open intervals, so by proposition 2.1 a set is open in $\mathbb{R}$ iff it is a union of open intervals:

$$\tau_{\mathbb{R}} = \left\{ \bigcup_{\alpha \in A} (a_\alpha, b_\alpha) \mid (a_\alpha, b_\alpha) \subseteq \mathbb{R} \text{ open intervals} \right\}.$$

In the discrete metric every singleton $\{x\} = B_{1/2}(x)$ is an open ball, so every subset of X is open.

Definition 2.5 (continuity at a point). A function $f : X \rightarrow Y$ between metric spaces (X, d_X) and (Y, d_Y) is *continuous at* $x \in X$ if for every $\varepsilon > 0$ there is some $\delta > 0$ such that

$$f(B_\delta(x)) \subseteq B_\varepsilon(f(x)),$$

i.e., $d_X(x, x') < \delta$ implies $d_Y(f(x), f(x')) < \varepsilon$. The function f is *continuous* if it is continuous at every point of X .

Theorem 2.1. *A function $f : X \rightarrow Y$ between metric spaces is continuous at $x \in X$ iff for every neighborhood V of $f(x)$ there is a neighborhood U of x with $U \subseteq f^{-1}(V)$, equivalently with $f(U) \subseteq V$.*